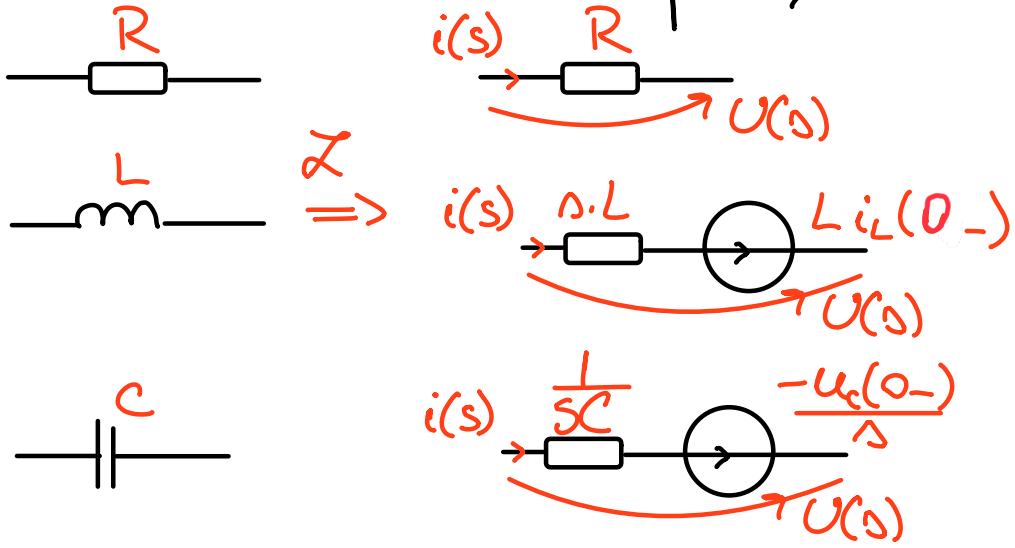
Circuite electrice în regim tranzitoriu

Algoritm

→ condiții inițiale ($t<0$)

$u_c(0_-)$; $i_L(0_-)$

schemă echivalentă în operațional ($t>0$)

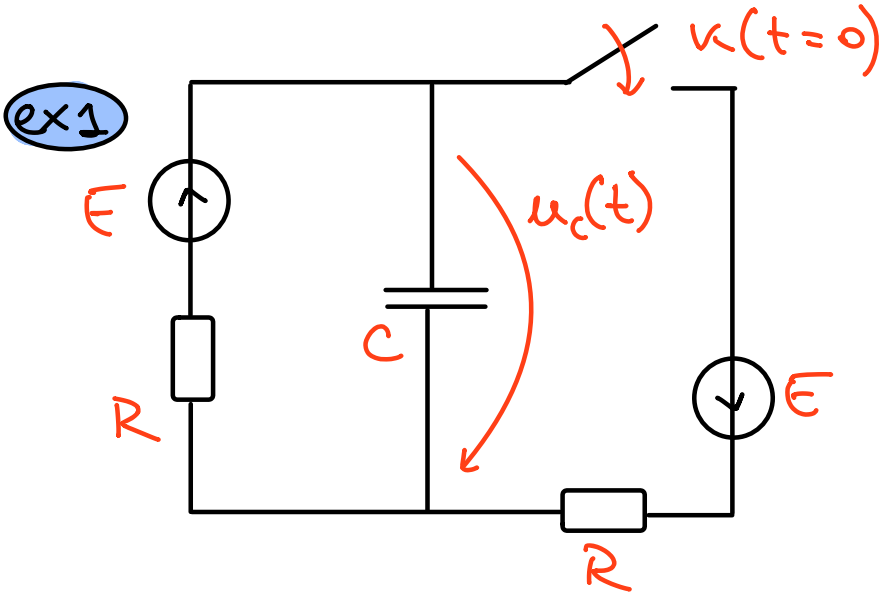


metoda de analiză curenți ochiuri.

aflați $Y(s)$ $U(s)$

transformata Laplace inversă

$i(t), u(t)$



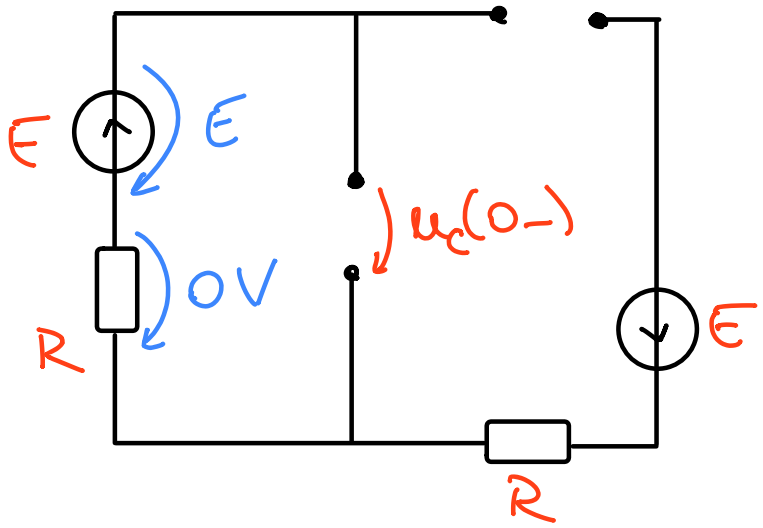
$E = 12V$

$R = 1k\Omega$

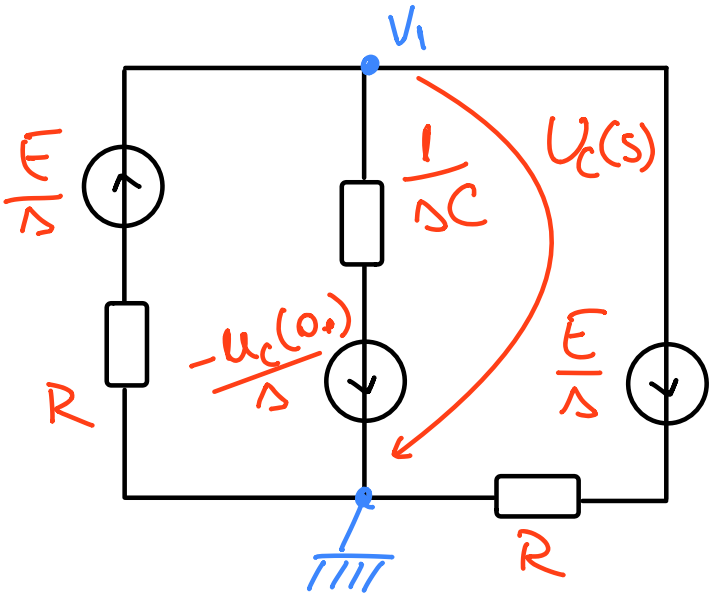
$C = 1mF$

$u_c(t) = ?$

cond. initiale



$$u_c(0_-) = E = 12\,V$$



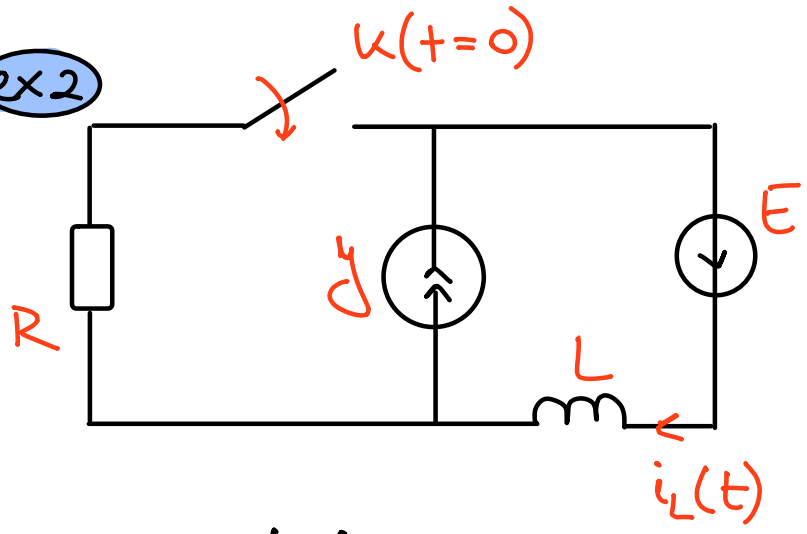
$$V_1\left(\frac{1}{R} + \frac{1}{1/(sC)} + \frac{1}{R}\right) = \frac{\frac{E}{s}}{R} - \frac{-\frac{u_c(0_-)}{s}}{\frac{1}{sC}} - \frac{\frac{E}{s}}{R} \Rightarrow$$

$$\Rightarrow V_1\left(\frac{2}{10^3} + s\cdot 10^{-3}\right) = 12\cdot 10^{-3} \Rightarrow V_1 = \frac{12}{2+s} = U_c(s)$$

$$F(s) = \frac{M(s)}{N(s)} \Rightarrow f(t) = \sum_{k=1}^{n} \frac{M(s_k)}{N'(s_k)} e^{s_k t} \qquad s_k \text{ rad a lui } N = 0$$

$$u_c(t) = \frac{12}{1}\cdot e^{-2t} \xrightarrow{\text{verificare}} \begin{array}{l} t=0 \Rightarrow u_c = 12 \\ t=\infty \Rightarrow u_c = 0\,V \end{array}$$

ex2



$R = 3\,\Omega$

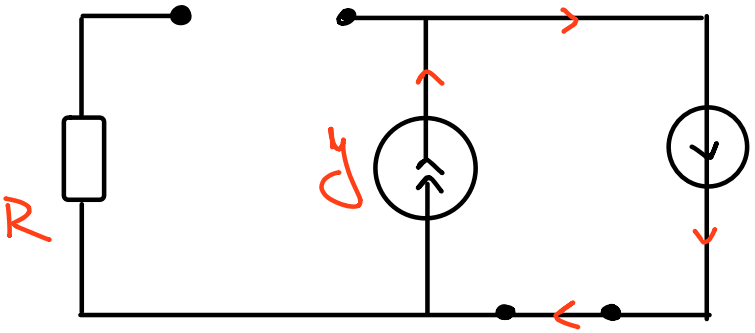
$L = 300\ mH$

$J = 1\,A$

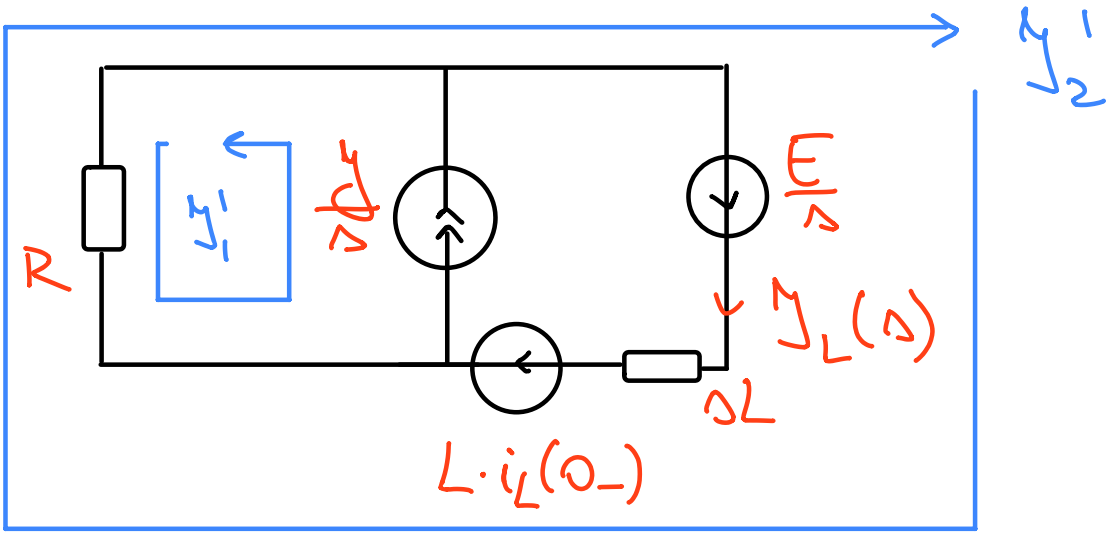
$E = 12\,V$

$i_L(t) = ?$

cond. initiale



$i_L(0_-) = J = 1\,A$



$$J_1' = \frac{\frac{J}{s}}{s} = \frac{1}{s}$$

$$sL \cdot J_2' - L \cdot i_L(0_-) + R\left(-J_1' + J_2'\right) - \frac{E}{s} = 0$$

$$s \cdot 0{,}3 \cdot J_2' - 0{,}3 \cdot 1 + 3\left(-\frac{1}{s} + J_2'\right) - \frac{12}{s} = 0$$

$$J_2'\left(3 + 0{,}3s\right) = 0{,}3 + \frac{12}{s} + \frac{3}{s} \Rightarrow J_2' = \frac{\frac{15}{s} + 0{,}3}{3 + 0{,}3s}$$

$$Y_2'(s) = \frac{15+0{,}3s}{s(3+0{,}3s)} = Y_L(s)$$

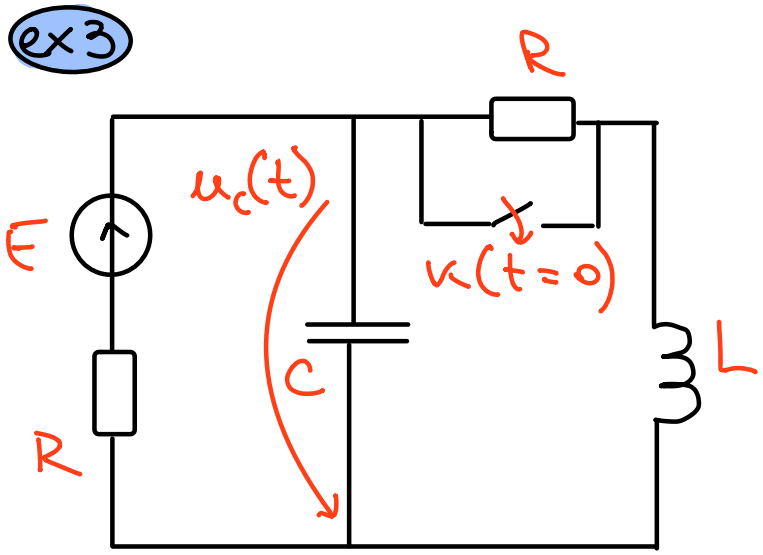
$$F(s) = \frac{M(s)}{s \cdot P(s)} \Rightarrow f(t) = \frac{M(0)}{P(0)} + \sum_{k=1}^{P} \frac{M(s_k)}{s_k \cdot P'(s_k)} e^{s_k t} \qquad s_k \to P = 0$$

$$i_L(t) = \frac{15}{3} + \frac{12}{-10 \cdot 0{,}3} e^{-10t} = 5 - 4e^{-10t}$$

verificare → $t = 0 \to i = 5 - 4 = 1\,A$

$t = \infty \to i = 5\,A$

ex 3



$R = 1\,\Omega$

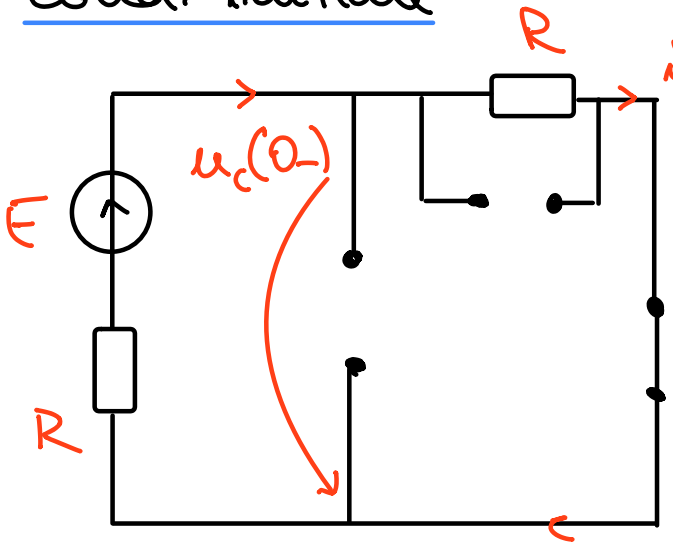
$L = \frac{3}{2}\,H$

$C = \frac{1}{3}\,F$

$E = 6\,V$

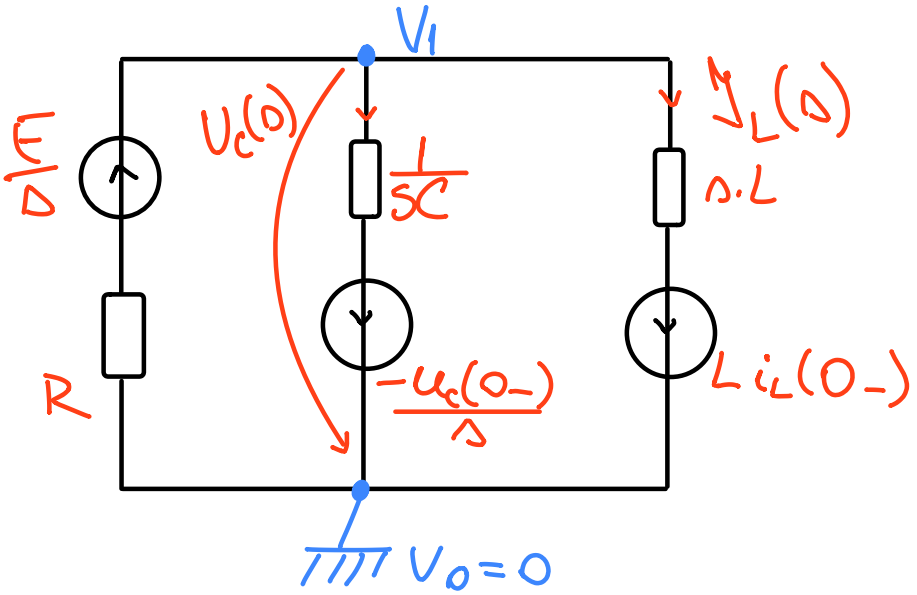
$u_c(t) = ?$

cond. inițiale



$$i_L(0_-) = \frac{E}{2R} = \frac{6}{2} = 3A$$

$$u_c(0_-) = i_L(0_-)R = 3 \cdot 1 = 3V$$



$$V_1\left(\frac{1}{R}+\frac{1}{\frac{1}{sC}}+\frac{1}{sL}\right)=\frac{\frac{E}{s}}{R}-\frac{\frac{-u_c(0_-)}{s}}{\frac{1}{sC}}-\frac{L\, i_L(0_-)}{s\cdot L}$$

$$V_1\left(1+\frac{s}{3}+\frac{2}{3s}\right)=\frac{6}{s}+\frac{3}{3}-\frac{3}{s}$$

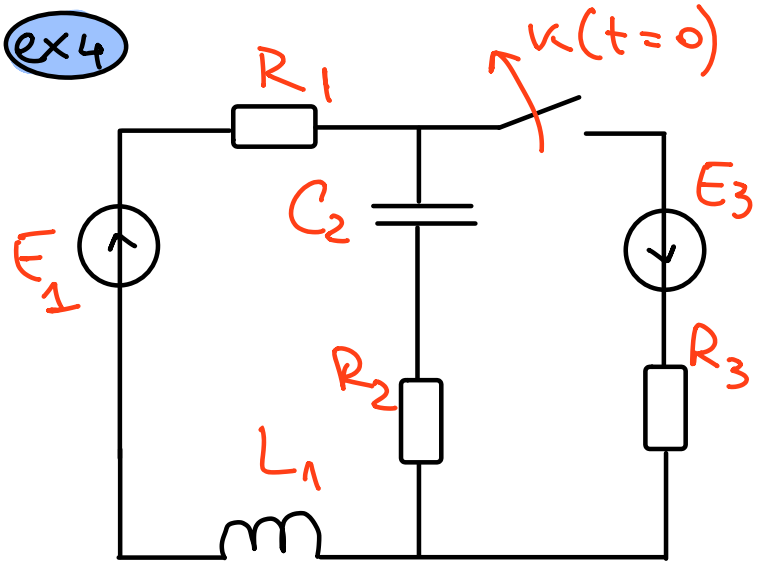
$$V_1\frac{s^2+3s+2}{3s}=\frac{3}{s}+1=\frac{3+s}{s}\Rightarrow V_1=\frac{9+3s}{s^2+s+2}=U_c(s)$$

$$u_c(t)=\frac{9+3(-1)}{2(-1)+3}\cdot e^{-t}+\frac{9+3(-2)}{2(-2)+3}e^{-2t}=6e^{-t}-3e^{-2t}$$

verific. $\rightarrow$ $t=0\Rightarrow u_c(0)=6-3=3$

$t=\infty\Rightarrow u_c(0)=0$

ex4



$E_1 = 100V$

$E_3 = 50V$

$R_1 = R_2 = 15\,\Omega$

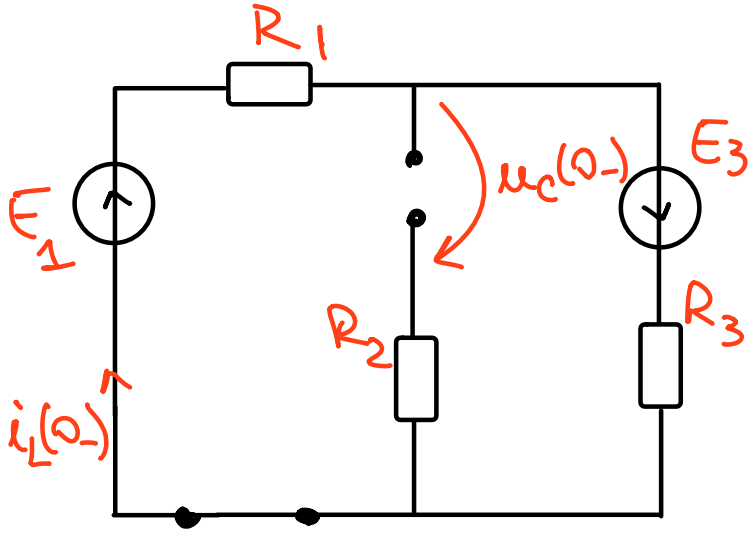
$R_3 = 10\,\Omega$

$C_2 = 5mF$

$L_1 = 1H$

$i_L(t) = ?$

cond. initiale

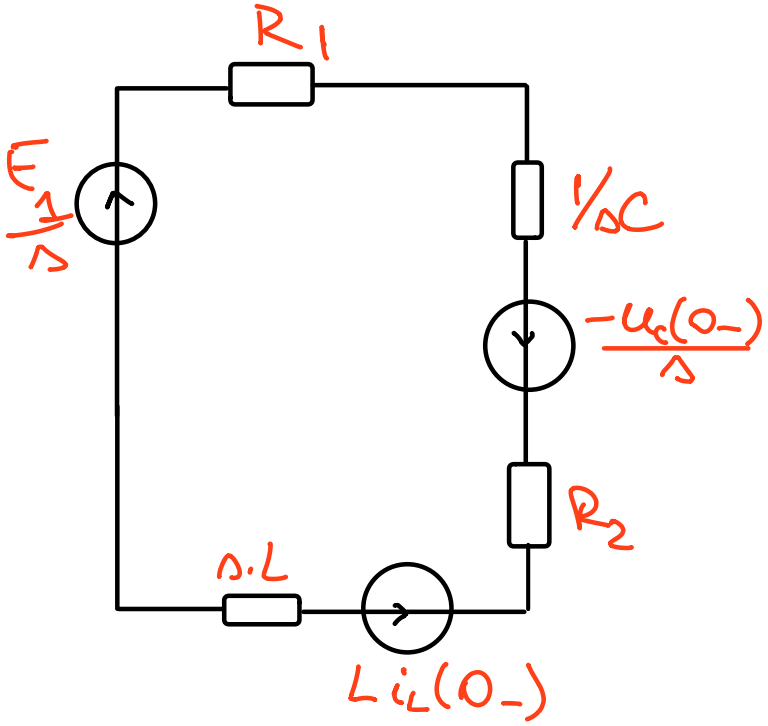


$$i_L(0_-) = \frac{E_1 + E_3}{R_1 + R_3}$$

$$i_L(0_-) = \frac{150}{25} = 6A$$

$$u_c(0_-) = -E_3 + i_L(0_-)R_3$$

$$= -50 + 6 \cdot 10 = 10\,V$$



$$Y_L(s)\left(R_1 + \frac{1}{sC} + R_2 + sL\right) = \frac{E_1}{s} - \frac{u_c(0_-)}{s} + L i_L(0_-)$$

$$I_L(s)\left(s+30+\frac{200}{s}\right)=\frac{100}{s}-\frac{10}{s}+6$$

$$I_L(s)=\frac{90+6s}{s^2+30s+200}\Rightarrow i_L(t)=\frac{30}{10}e^{-10t}+\frac{-30}{-10}e^{-20t}$$

$$i_L(t)=3e^{-10t}+3e^{-20t}$$